Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Lecture 5: BVAR Extensions

Hedibert Freitas Lopes

The University of Chicago Booth School of Business
5807 South Woodlawn Avenue, Chicago, IL 60637
http://faculty.chicagobooth.edu/hedibert.lopes

hlopes@ChicagoBooth.edu

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Outline

1. Variable selection
   - ON/OFF switch
   - SSVS
2. Factor analysis
   - Basic model
   - PBFA
   - FSV
   - SDFM
3. Forecasting with large $q$
4. Forecasting with many predictors
5. Large BVAR & BFAVAR
6. Large TVP-BVAR & BFAVAR
7. References

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Why variable selection in VAR?

VAR flexibility $\Rightarrow$ over-parameterization $\Rightarrow$ poor forecasts.

First solution: Minnesota prior of Doan *et al.* (1984).

Additional (more recent) solutions:

- Factor models (Stock and Watson, 2006)
- Variable selection priors (George *et al.*, 2008)
- Steady-state priors (Villani, 2009)
- Bayesian model averaging (Garratt *et al.*, 2009)
- Combination forecasts (Clark and McCracken, 2010)

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Stochastic search

According to Korobilis (2011) "stochastic search"

> *. . . simply means that if the model space is too large to assess in a deterministic manner (i.e. enumerate and estimate all possible models, and decide on the best one using some goodness-of-fit measure), the algorithm will visit only the most probable models in a stochastic manner.*

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# ON/OFF switch

Recall the basic VAR(p) model

$$y_t = B_1 y_{t-1} + \ldots + B_p y_{t-p} + u_t$$

where $u_t \sim$ i.i.d. $N(0, \Sigma)$ and $B = (B_1, \ldots, B_p)$.

Bayesian variable selection idea

$$B_{ij} = \begin{cases} B_{ij} = 0 & \gamma_{ij} = 0 \\ B_{ij} \neq 0 & \gamma_{ij} = 1 \end{cases}$$

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Constrained VAR

The VAR(p) model can be rewritten as a SUR model as

$$y_t = z_t\beta + u_t$$

where $z_t = I_q \otimes (y'_{t-1}, \ldots, y'_{t-p})$ is $(q \times s)$, $\beta = \text{vec}(B)$ is $(s \times 1)$ and $s = q^2 p$.

Including the variable selection indicators:

$$y_t = z_t\Gamma\beta + u_t$$

where $\Gamma = \text{diag}(\gamma_{11}, \ldots, \gamma_{1,pq}, \ldots, \gamma_{q1}, \ldots, \gamma_{q,pq})$.

$2^s$ VAR models $\Rightarrow$ variable selection $=$ model selection.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Prior specification

VAR coefficients

$$\beta \sim N(\beta_0, V_0)$$

VAR error variance

$$p(\Sigma) \propto |\Sigma|^{(q+1)/2}$$

Variable selection indicators

$$\gamma_j | \gamma_{-j} \sim Bernoulli(\pi_{0j})$$

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Full conditional of $\beta$

Let $\tilde{z}_t = z_t\Gamma$. Then,

$$p(\beta|\Sigma,\gamma,y) \propto \exp\left\{-0.5\left[\beta'(V_0+A_1)^{-1}\beta - 2\beta'(V_0^{-1}\beta_0 + A_2)\right]\right\}$$

where

$$A_1 = \sum_{t=1}^{T} \tilde{z}_t'\Sigma^{-1}\tilde{z}_t \quad \text{and} \quad A_2 = \sum_{t=1}^{T} \tilde{z}_t'\Sigma^{-1}y_t.$$

Therefore,

$$\beta|\Sigma,\gamma,y \sim N(\beta_1, V_1)$$

where

$$\beta_1 = V_1(V_0^{-1}\beta_0 + A_2) \quad \text{and} \quad V_1^{-1} = V_0^{-1} + A_1$$

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Full conditional of $\Sigma$

$$p(\Sigma|\beta, y) \propto |\Sigma|^{\frac{q+T+1}{2}} \exp\left\{-0.5 \sum_{t=1}^{T} (y_t - z_t\beta)'\Sigma^{-1}(y_t - z_t\beta)\right\}$$

or

$$\Sigma|\beta, y \sim IW(T, S)$$

where

$$S = \sum_{t=1}^{T} (y_t - z_t\beta)(y_t - z_t\beta)'$$

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Full conditional of $\gamma$

It is easy to see that

$$p(\gamma_j = 1|\gamma_{-j}, \beta, \Sigma, y) \propto p(y|\theta_j, \Sigma, \gamma_{-j}, \gamma_j = 1)\pi_{0j} = A_j$$

and

$$p(\gamma_j = 0|\gamma_{-j}, \beta, \Sigma, y) \propto p(y|\theta_j, \Sigma, \gamma_{-j}, \gamma_j = 1)(1 - \pi_{0j}) = B_j$$

Therefore

$$(\gamma_j|\gamma_{-j}, \beta, \Sigma, y) \sim \text{Bernoulli}\left(\frac{A_j}{A_j + B_j}\right)$$

Variable selection
ON/OFF switch
ssvs
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# SSVS

Jochmann *et al.* (2011) adapts George and McCulloch's (1993, 1997) stochastic search variable selection (SSVS) method to BVAR with structural breaks.

The spike-slab prior for $\gamma_j$ is given by

$$\beta_j|\gamma_j \sim (1-\gamma_j)N(0,\kappa_{0j}^2) + \gamma_j N(0,\kappa_{1j}^2).$$

SSVS aspect of the prior

small $\kappa_{0j}^2$ (coefficient is virtually zero)

large $\kappa_{1j}^2$ (relatively non-informative prior).

Indicator variable

$$\gamma_i \sim \text{Bernoulli}(q_j)$$

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Default semi-automatic approach

George *et al.* (2008) suggests

$$\begin{aligned} \kappa_{0j} &= c_0 \sqrt{\hat{\text{var}}(\beta_j)} \\ \kappa_{1j} &= c_1 \sqrt{\hat{\text{var}}(\beta_j)}, \end{aligned}$$

where $c_0 << c_1$.

$\hat{\text{var}}(\beta_j)$ is an estimate of the variance of the coefficient in an unrestricted VAR (e.g., the ordinary least squares quantity or an estimate based on a preliminary MCMC run of the VAR using a non-informative prior).
Jochmann *et al.* (2011) set $c_0 = 0.1$ and $c_1 = 10$.

George (2000) reviews the variable selection problem.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Prior on $\Sigma$

Let $\Psi$ be the lower-triangular Cholesky root of $\Sigma^{-1}$, i.e.

$$\Sigma^{-1} = \Psi\Psi'.$$

## Diagonal components

$$\psi_{jj}^2 \sim G(a_j, b_j)$$

## Off-diagonal components

$$\begin{aligned} \psi_{ij} &\sim (1-\omega_{ij})N(0,\kappa_{0ij}^2) + \omega_{ij}N(0,\kappa_{1ij}^2) \\ \omega_{ij} &\sim \text{Bernoulli}(q_{ij}) \end{aligned}$$

Substantial progress has been made on Bayesian modeling by imposing restrictions on the $\Sigma$ matrix (Pourahmadi, 1999, Sims and Zha, 1998, Smith and Kohn, 2002, Waggoner and Zha, 2003, Lopes, McCulloch and Tsay, 2012, Lopes and Polson, 2012).

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Bayesian factor analysis (BFA)[1]

- Pre-MCMC era
  - Press (1972)
  - Martin and McDonald (1975)
  - Bartholomew (1981, 1987)
  - Press and Shigemasu (1989)
- MCMC era
  - Geweke and Zhou (1996)
  - Aguilar and West (2000)
  - Lopes (2000)
  - Lopes and West (2004)
  - West (2003)
  - Lopes and Carvalho (2007)
  - Lopes, Salazar and Gamerman (2008)

[1] Lopes (2003) Factor models. *ISBA Bulletin*, 10(2), 7-10.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Basic model

For any specified positive integer $k \leq m$, the standard $k-$factor model relates each $y_t$ to an underlying $k-$vector of random variables $f_t$, the common factors, via

$$\begin{aligned} y_t|f_t &\sim N(\beta f_t, \Sigma) \\ f_t &\sim N(0, I_k) \end{aligned}$$

where $\Sigma = \text{diag}(\sigma_1^2, \cdots, \sigma_m^2)$. Therefore,

- $\text{var}(y_{it}|f) = \sigma_i^2$,
- $\text{cov}(y_{it}, y_{jt}|f) = 0$,
- $V(y_t|\beta, \Sigma) = \Omega = \beta\beta' + \Sigma$
- $\text{var}(y_{it}) = \sum_{l=1}^{k} \beta_{il}^2 + \sigma_i^2$,
- $\text{cov}(y_{it}, y_{jt}) = \sum_{l=1}^{k} \beta_{il}\beta_{jl}$.

Common factors explain all the dependence structure among the $m$ variables.

Variable selection
ON/OFF switch SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Invariance

The model is invariant under transformations of the form $\beta^* = \beta P'$ and $f_t^* = P f_t$, where $P$ is any orthogonal $k \times k$ matrix.

Classical approach: $\beta' \Sigma^{-1} \beta = I$.

Our approach: $\beta$ is a block lower triangular.

$$\beta = \begin{pmatrix} \beta_{11} & 0 & 0 & \cdots & 0 \\ \beta_{21} & \beta_{22} & 0 & \cdots & 0 \\ \beta_{31} & \beta_{32} & \beta_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \beta_{k1} & \beta_{k2} & \beta_{k3} & \cdots & \beta_{kk} \\ \beta_{k+1,1} & \beta_{k+1,2} & \beta_{k+1,3} & \cdots & \beta_{k+1,k} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \beta_{m1} & \beta_{m2} & \beta_{m3} & \cdots & \beta_{mk} \end{pmatrix}$$

This form provides both identification and, often, useful interpretation of the factor model.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Number of parameters

The resulting factor form of $\Omega$ has

$$m(k+1) - k(k-1)/2$$

parameters, compared with the total

$$m(m+1)/2$$

in an unconstrained (or $k = m$) model, leading to the constraint that

$$m(m+1)/2 - m(k+1) + k(k-1)/2 \geq 0,$$

which provides an upper bound on $k$.

For example,

- $m = 6$ implies $k \leq 3$,
- $m = 12$ implies $k \leq 7$,
- $m = 20$ implies $k \leq 14$,
- $m = 50$ implies $k \leq 40$,

Even for small $m$, the bound will often not matter as relevant $k$ values will not be so large.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Full-rank loading matrix

Geweke and Singleton (1980) show that, if $\beta$ has rank $r < k$ there exists a matrix $Q$ such that $\beta Q = 0$ and $Q'Q = I$ and, for any orthogonal matrix $M$,

$$\beta\beta' + \Sigma = (\beta + MQ')'(\beta + MQ') + (\Sigma - MM')$$

This translation invariance of $\Omega$ under the factor model implies lack of identification and, in application, induces symmetries and potential multimodalities in resulting likelihood functions.

This issue relates intimately to the question of uncertainty of the number of factors.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Ordering of the variables

Alternative orderings are trivially produced via $Ay_t$ for some rotation matrix $A$.

The new rotation has the same latent factors but transformed loadings matrix $A\beta$.

$$Ay_t = A\beta f + \varepsilon_t$$

This new loadings matrix does not have the lower triangular structure.

However, we can always find an orthonormal matrix $P$ such that $A\beta P'$ is lower triangular, and so simply recover the factor model with the same probability structure for the underlying latent factors $Pf_t$ (Lopes and West, 2004).

The order of the variables in $y_t$ is irrelevant assuming that $k$ is properly chosen.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Prior specification

Loading matrix:

$$\begin{aligned} \beta_{ij} &\sim N(0, C_0) && \text{when } i \neq j, \\ \beta_{ii} &\sim N(0, C_0)1(\beta_{ii} > 0) && \text{when } i = 1, \ldots, k \end{aligned}$$

Idiosyncratic variances

$$\sigma_i^2 \sim IG(\nu/2, \nu s^2/2)$$

where $s^2$ is the prior mode of each $\sigma_i^2$ and $\nu$ the prior degrees of freedom hyperparameter.

We eschew the use of standard improper reference priors $p(\sigma_i^2) \propto 1/\sigma_i^2$, since such priors lead to the Bayesian analogue of the so-called *Heywood problem* (Martin and McDonald, 1975, and Ihara and Kano, 1995).

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Full conditional distributions

Factor scores

$$f_t \sim N(V_f \beta' \Sigma^{-1} y_t, V_f)$$

where $V_f = (I_k + \beta' \Sigma^{-1} \beta)^{-1}$.

Idiosyncrasies

$$\sigma_i^2 \sim IG((\nu + T)/2, (\nu s^2 + d_i)/2)$$

where $d_i = (y_i - f\beta_i')'(y_i - f\beta_i')$.

First $k$ rows of $\beta$

$$\beta_i \sim N(M_i, C_i) 1(\beta_{ii} > 0)$$

where

$$\begin{aligned} M_i &= C_i \left( C_0^{-1} \mu_0 1_i + \sigma_i^{-2} f_i' y_i \right) \\ C_i^{-1} &= C_0^{-1} I_i + \sigma_i^{-2} f_i' f_i. \end{aligned}$$

Last $m - k$ rows of $\beta$

$$\beta_i \sim N(M_i, C_i)$$

where

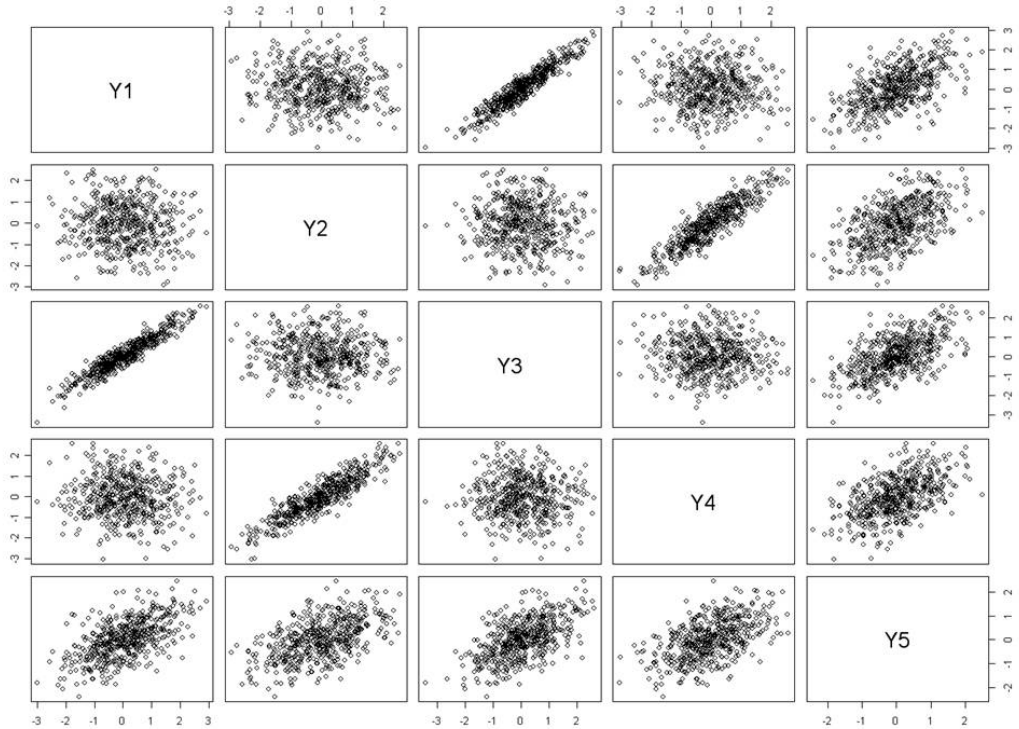
$$\begin{aligned} M_i &= C_i \left( C_0^{-1} \mu_0 1_k + \sigma_i^{-2} f' y_i \right) \\ C_i^{-1} &= C_0^{-1} I_k + \sigma_i^{-2} f' f. \end{aligned}$$

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Simulation exercise

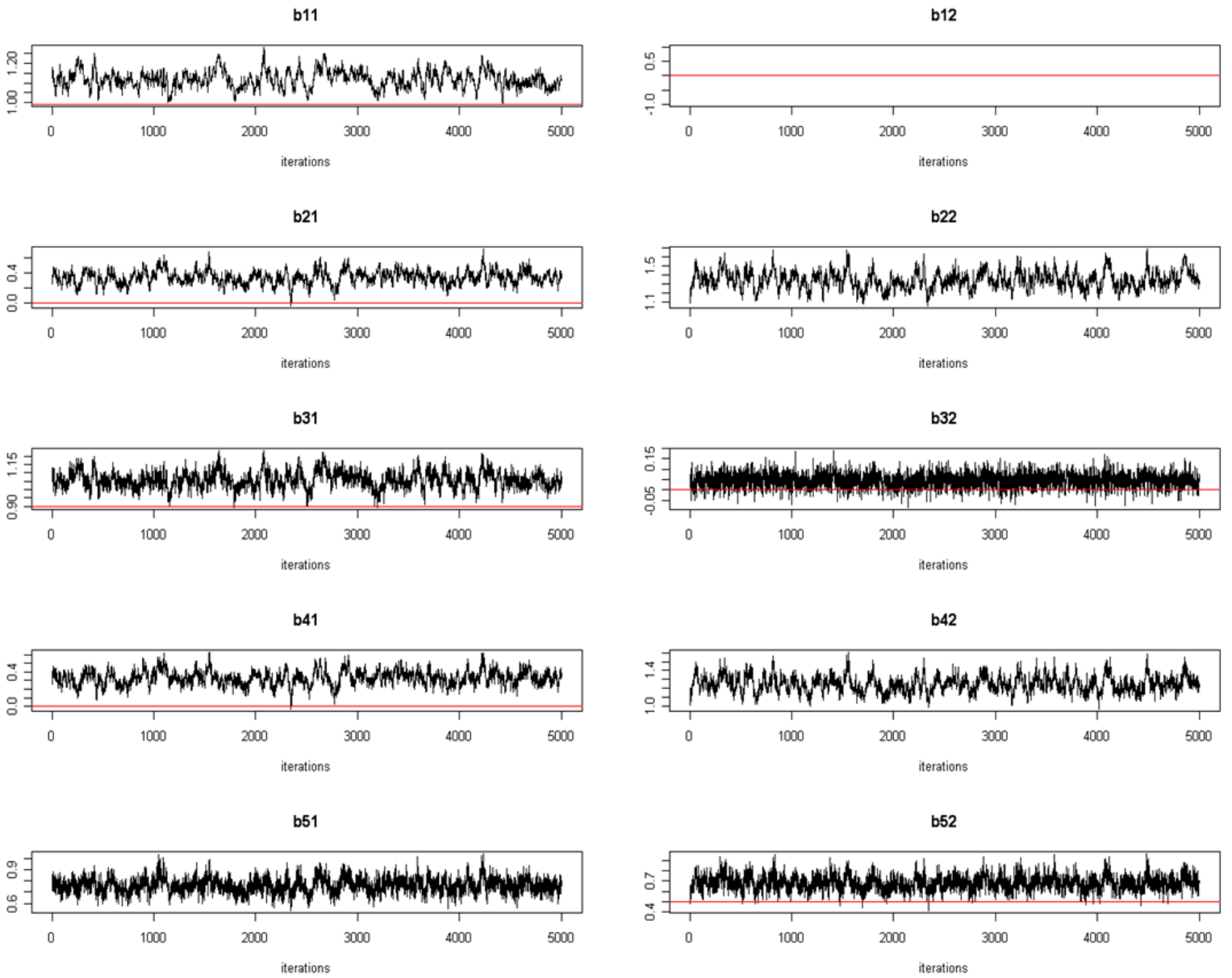
$T = 500$, $p = 5$ and $k = 2$.

$$\beta = \begin{pmatrix} 0.99 & 0.00 \\ 0.00 & 0.99 \\ 0.90 & 0.00 \\ 0.00 & 0.90 \\ 0.50 & 0.50 \end{pmatrix} \qquad \Sigma = \text{diag} \begin{pmatrix} 0.01 \\ 0.05 \\ 0.10 \\ 0.15 \\ 0.20 \end{pmatrix}$$



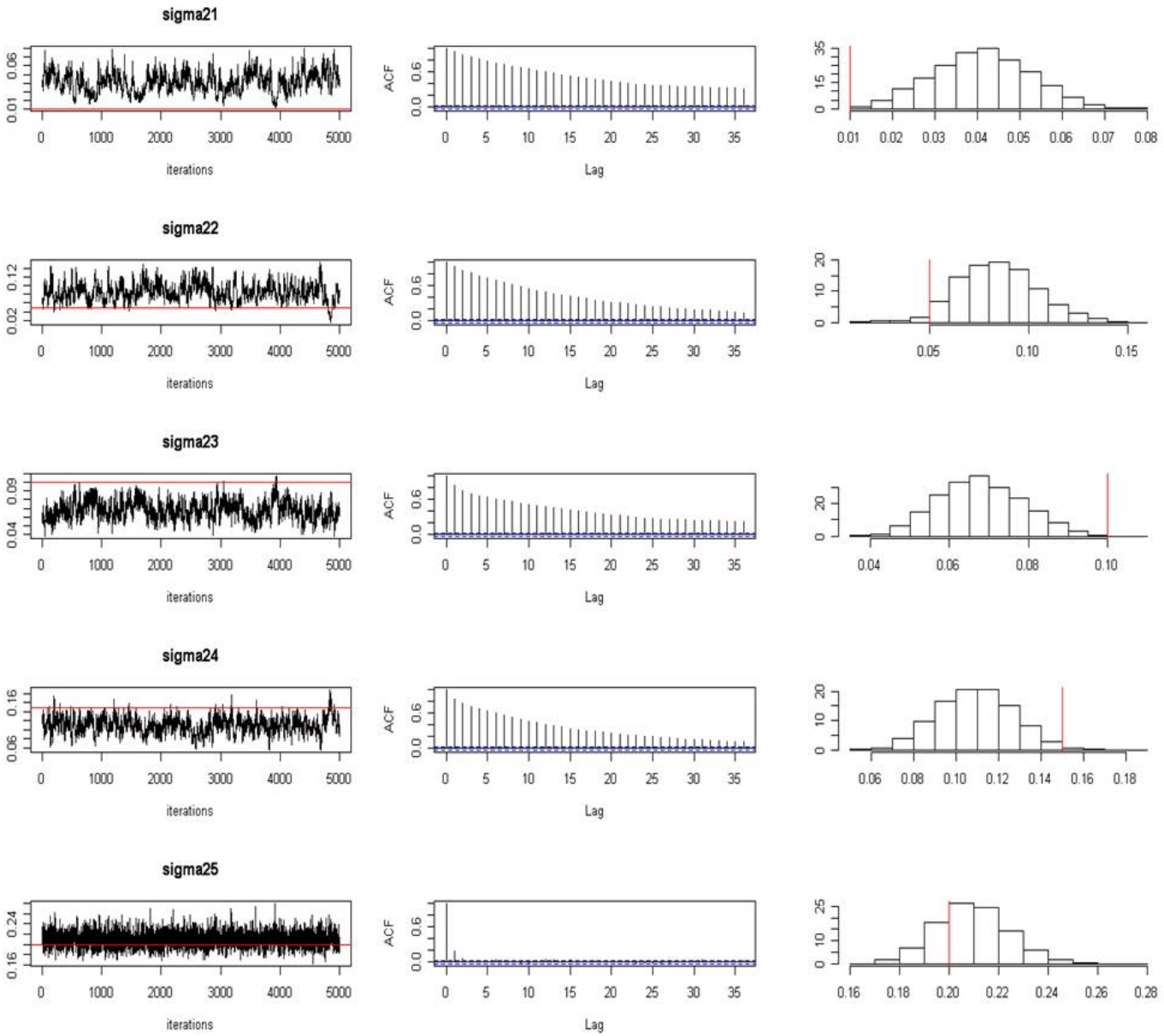
Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Factor loadings



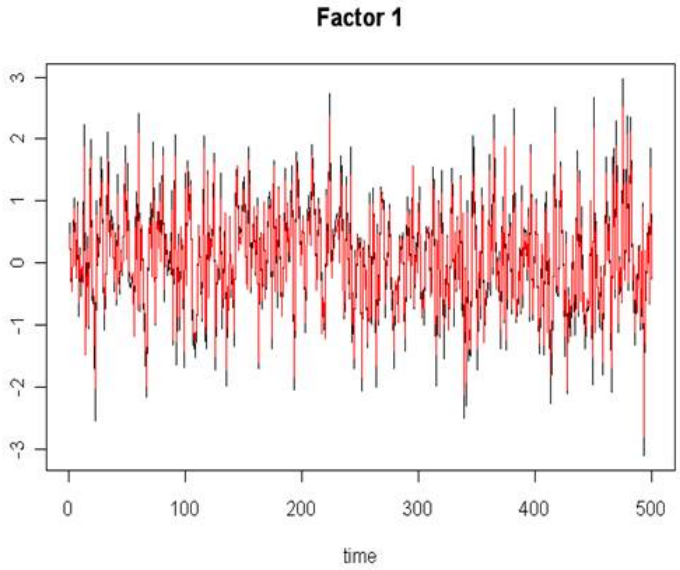
Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

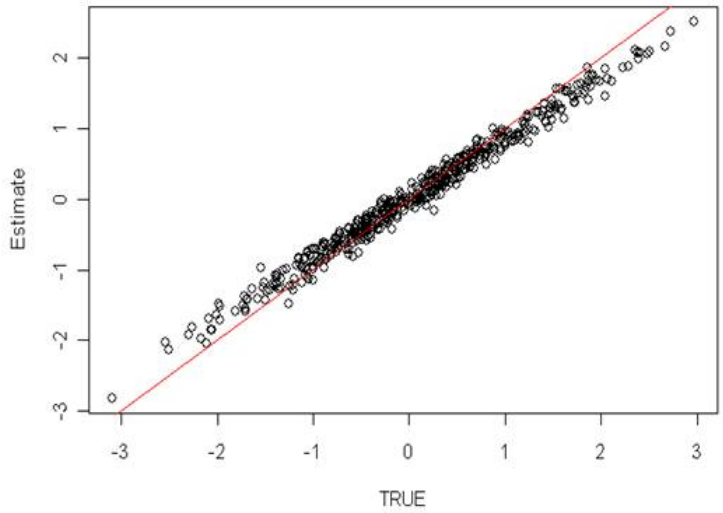
# Idiosyncratic variances

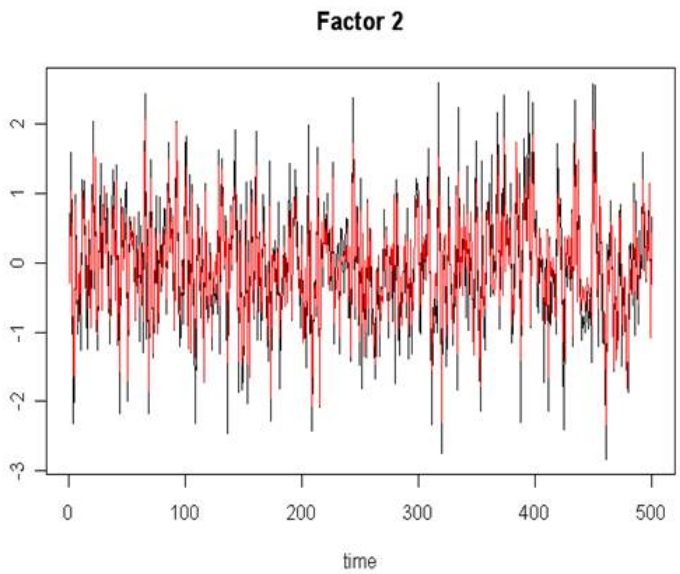


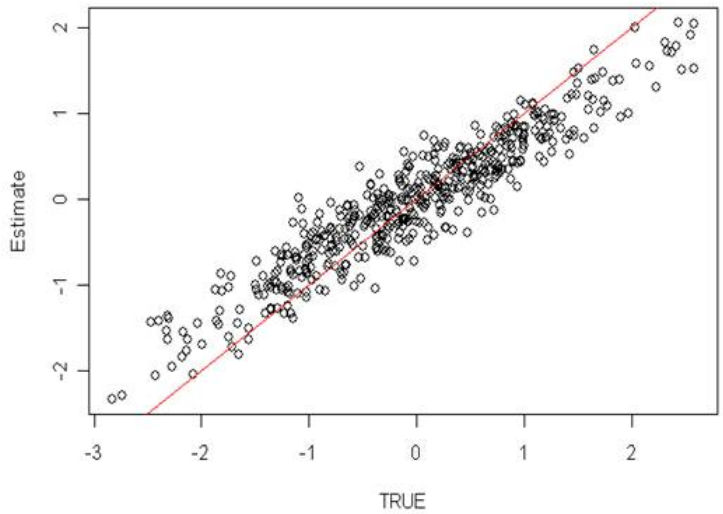
Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Common factors



Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Exchange rates[2]

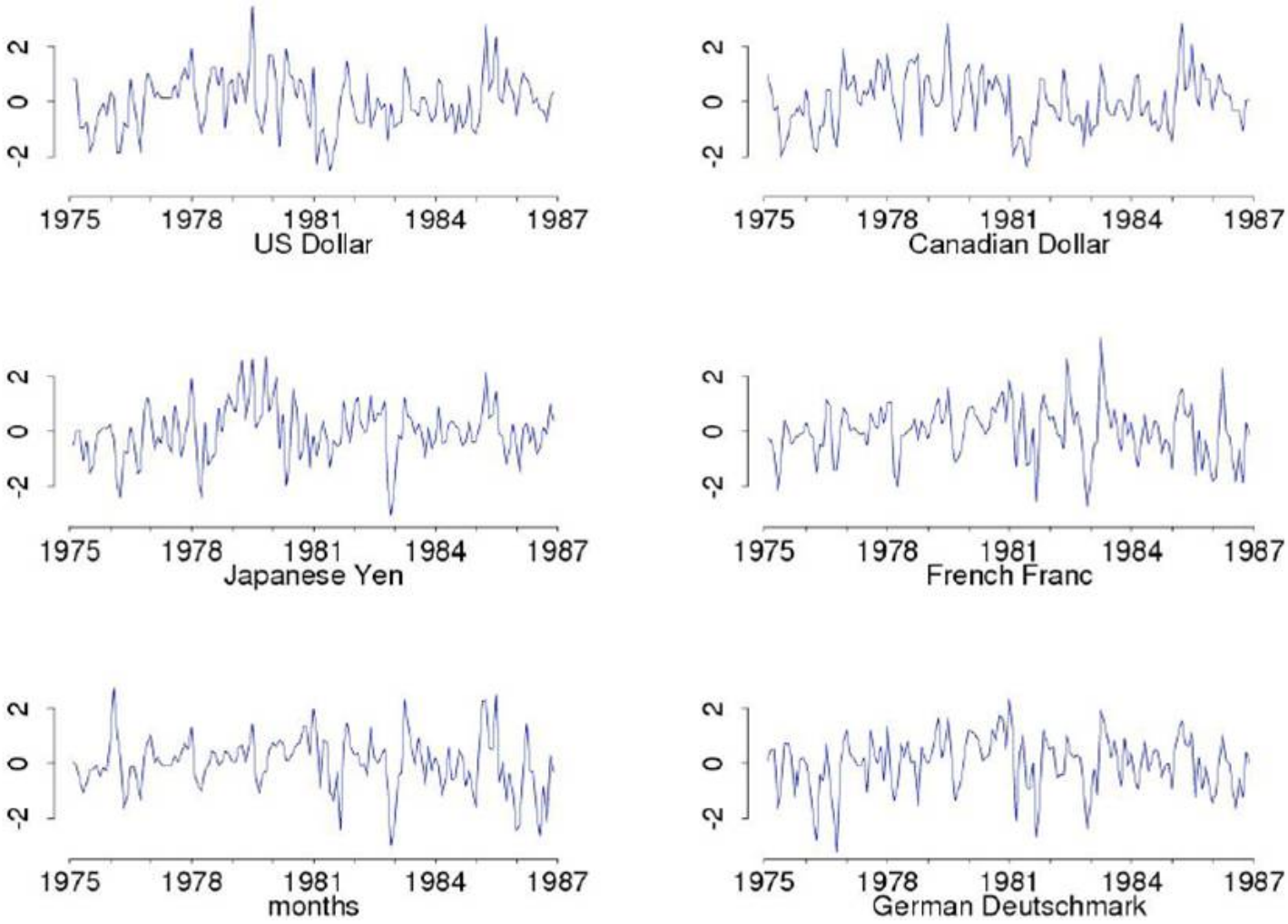
- Monthly international exchange rates.
- The data span the period from 1/1975 to 12/1986.
- Time series are the exchange rates in British pounds of
  - US dollar (US)
  - Canadian dollar (CAN)
  - Japanese yen (JAP)
  - French franc (FRA)
  - Italian lira (ITA)
  - (West) German (Deutsch)mark (GER)

[2] Lopes and West (2004)

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Data

## Standardized first differences of monthly log exchange rates




Standardized first differences of monthly observed exchange rates.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Posterior means

## 1st ordering

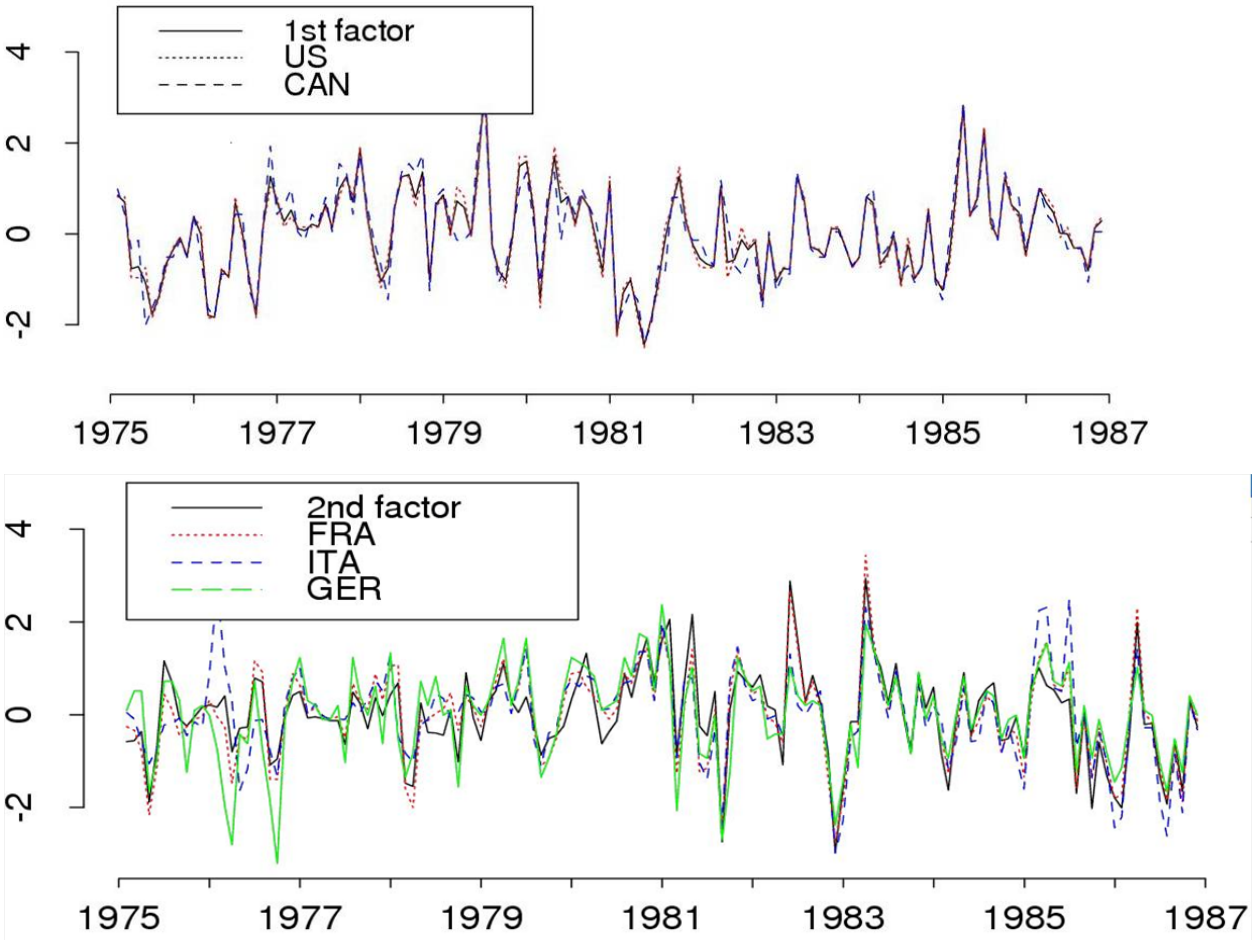
$$E(\beta|y) = \begin{pmatrix} \text{US} & 0.99 & 0.00 \\ \text{CAN} & 0.95 & 0.05 \\ \text{JAP} & 0.46 & 0.42 \\ \text{FRA} & 0.39 & 0.91 \\ \text{ITA} & 0.41 & 0.77 \\ \text{GER} & 0.40 & 0.77 \end{pmatrix} \qquad E(\Sigma|y) = \text{diag} \begin{pmatrix} 0.05 \\ 0.13 \\ 0.62 \\ 0.04 \\ 0.25 \\ 0.28 \end{pmatrix}$$

## 2nd ordering

$$E(\beta|y) = \begin{pmatrix} \text{US} & 0.98 & 0.00 \\ \text{JAP} & 0.45 & 0.42 \\ \text{CAN} & 0.95 & 0.03 \\ \text{FRA} & 0.39 & 0.91 \\ \text{ITA} & 0.41 & 0.77 \\ \text{GER} & 0.40 & 0.77 \end{pmatrix} \qquad E(\Sigma|y) = \text{diag} \begin{pmatrix} 0.06 \\ 0.62 \\ 0.12 \\ 0.04 \\ 0.25 \\ 0.26 \end{pmatrix}$$

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Common factors



Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Parsimonious BFA[3]

Frühwirth-Schnatter and Lopes (2009) revisited LW2004. We...

- Lay down a new and general set of identifiability conditions that handles the ordering problem present in most of the current literature,
- Introduce a new strategy for searching the space of parsimonious/sparse factor loading matrices,
- Designed a highly computationally efficient MCMC scheme for posterior inference which makes several improvements over the existing alternatives,

for the important class of Gaussian factor models.

[3] Frühwirth-Schnatter and Lopes (2009) Parsimonious Bayesian factor analysis when number of factors is unknown. *Journal of Econometrics*.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Identification issues

- Block lower triangular
  Generalized lower triangular alternative

- Rank deficiency
  If $\beta$ is rank-deficient, then $\exists Q$ such that

$$\beta\beta' = (\beta + MQ')(\beta + MQ')' + (\Sigma - MM').$$

  for some orthogonal $M$ with $\beta Q = 0$ and $Q'Q = I$.
  We use this "deficiency" in our model search strategy.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Generalized lower triangular

$$\begin{pmatrix} \beta_{11} & 0 & 0 & 0 \\ \beta_{21} & \beta_{22} & 0 & 0 \\ \beta_{31} & \beta_{32} & \beta_{33} & 0 \\ \beta_{41} & \beta_{42} & \beta_{43} & \beta_{44} \\ \beta_{51} & \beta_{52} & \beta_{53} & \beta_{54} \\ \beta_{61} & \beta_{62} & \beta_{63} & \beta_{64} \\ \beta_{71} & \beta_{72} & \beta_{73} & \beta_{74} \end{pmatrix} \Longrightarrow \begin{pmatrix} \beta_{11} & 0 & 0 & 0 \\ \beta_{21} & 0 & 0 & 0 \\ \beta_{31} & \beta_{32} & 0 & 0 \\ \beta_{41} & \beta_{42} & 0 & 0 \\ \beta_{51} & \beta_{52} & 0 & 0 \\ \beta_{61} & \beta_{62} & 0 & \beta_{64} \\ \beta_{71} & \beta_{72} & 0 & \beta_{74} \end{pmatrix}$$

Birth/death of loadings
Birht/death of columns.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Other contributions

- Our approach provides a principled way for inference on the number of factors, as opposed to previous work (Carvalho et al., 2008; Bhattacharya and Dunson, 2009).
- Our prior specification on $\Sigma$ properly addresses Heywood problems
- Our fractional-like prior on $\beta$ is more robust than the existing ones (Lopes and West, 2004, Ghosh and Dunson, 2009)
- Efficient (and correct) parameter expansion where the prior is unchanged (as opposed to GD2009).

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# The British Cohort Study[4]

A survey of all babies born (alive or dead) after the 24th week of gestation from 00.01 hours on Sunday, 5th April to 24.00 hours on Saturday, 11 April, 1970 in England, Scotland, Wales and Northern Ireland.

Follow-ups (so far): 1975, 1980, 1986, 1996, 2000, 2004, 2008.

Background characteristics:

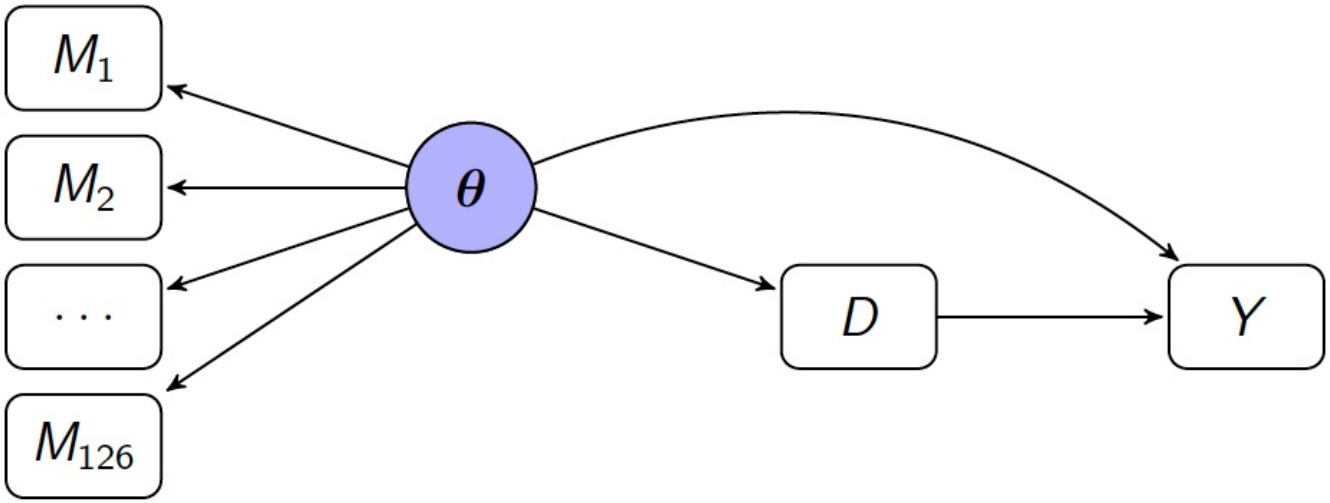
- Cognitive, mental, physical health measurements (age 10)
- Education and adult outcomes (age 30)

Sample size: 5,105 women and 5,420 men.

[4] Conti, Heckman, Frühwirth-Schnatter, Lopes and Piatek (2011) Constructing economically justified aggregates: an application to the early origins of health. *Journal of Econometrics*.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

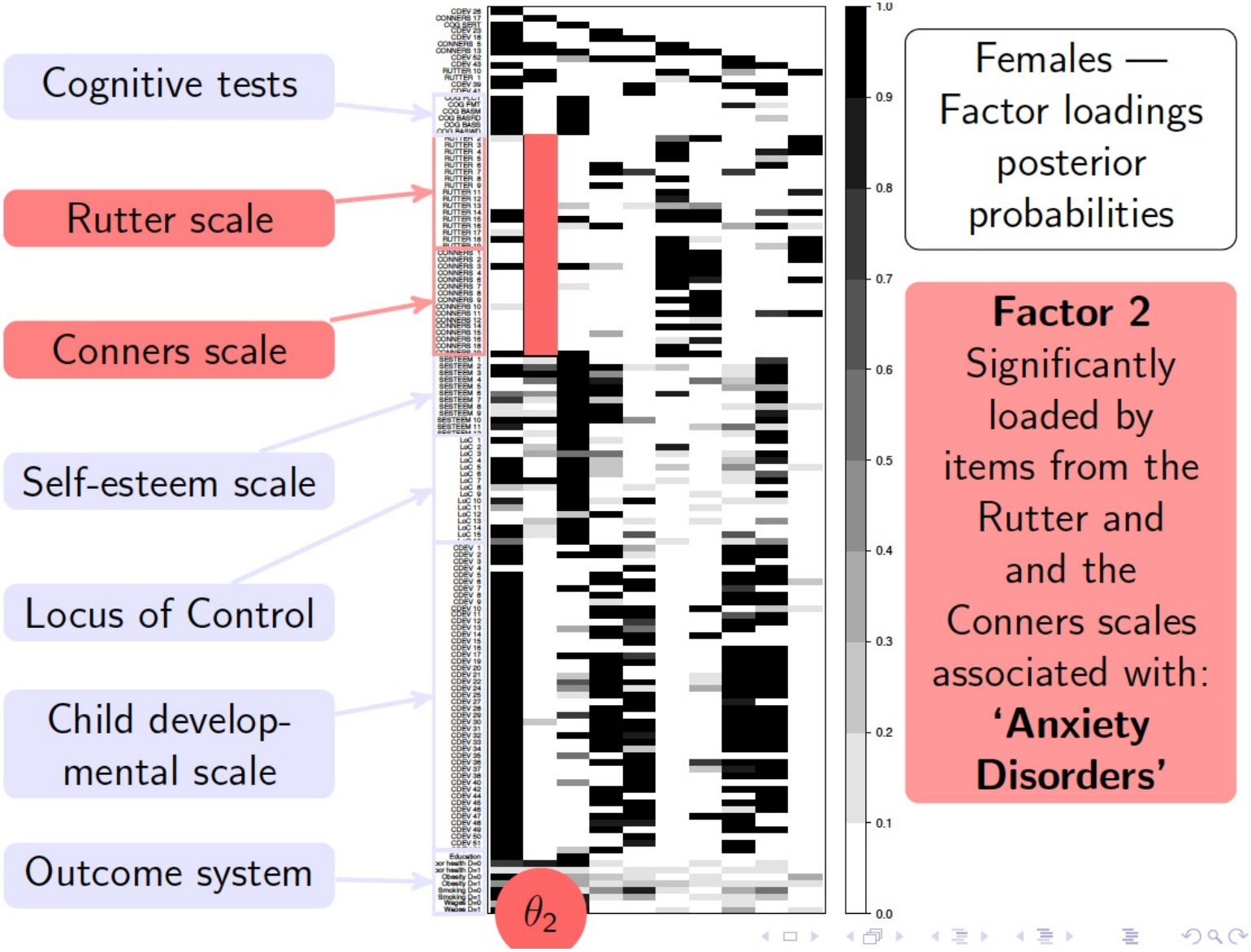
# Graphical model




| **Variables** | | **Definitions** |
|---|---|---|
| Education | $D$ | Observed: achieving A-level or higher |
| Outcomes | $Y$ | Observed in one state only!<br>Poor health, Obesity, Smoking, Wage |
| Measurements: | $M_j$ | Observed: 126 items (binary and cont.) |
| Cognitive skills<br>Personality | $\theta$ | **Unobserved dimensions** |

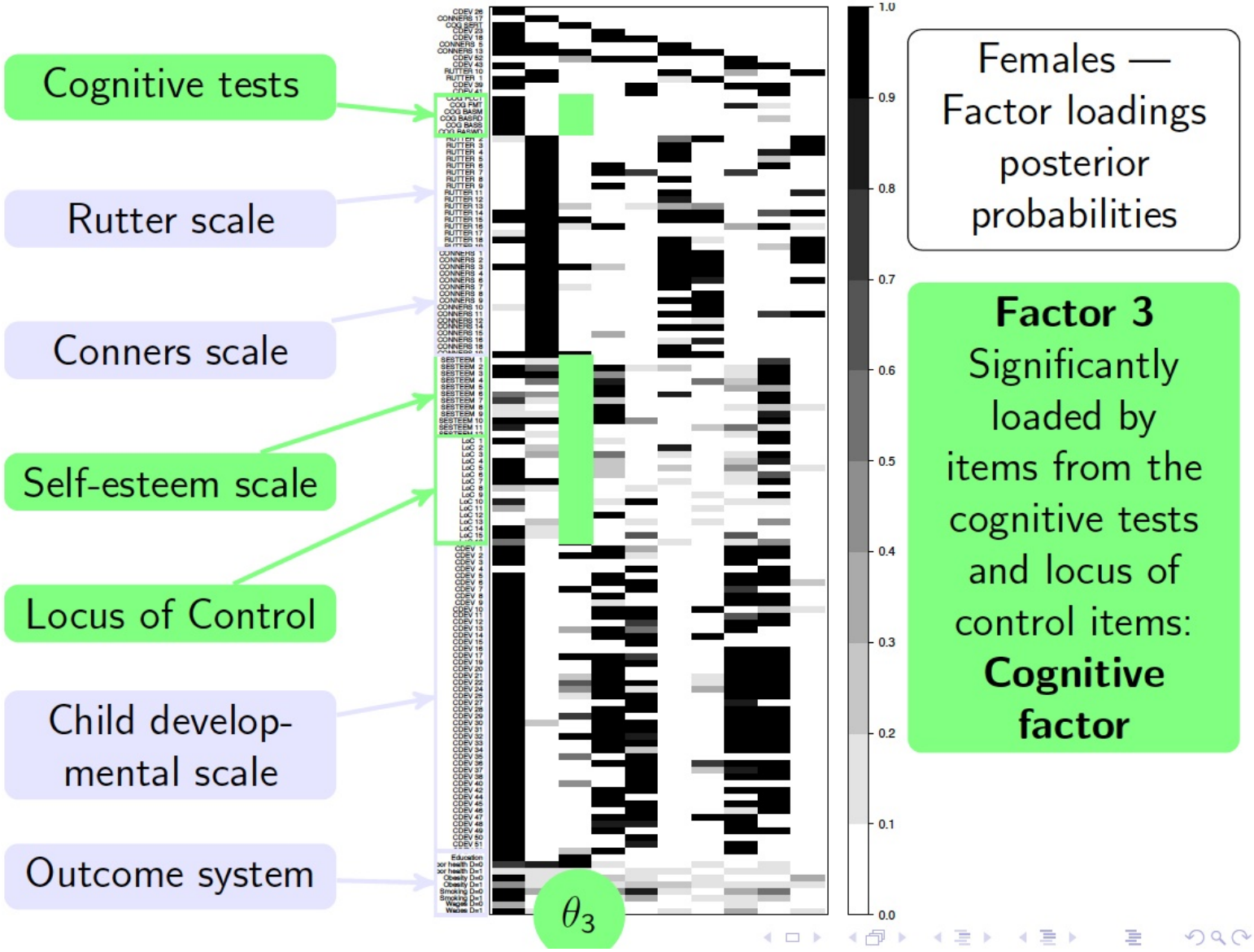
Variable selection
ON/OFF switch SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# 2nd factor



Variable selection
ON/OFF switch SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# 3rd factor



Variable selection
ON/OFF switch SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Factor stochastic volatility

Factor structure: $y_t$ follows $k$-order factor stochastic volatility model[5] when

$$\begin{aligned} y_t|f_t &\sim N(\beta f_t, \Sigma_t) \\ f_t &\sim N(0, H_t) \end{aligned}$$

and

$$\begin{aligned} \Sigma_t &= \text{diag}(\sigma_{1t}^2, \ldots, \sigma_{pt}^2) \\ H_t &= \text{diag}(\sigma_{p+1,t}^2, \ldots, \sigma_{p+k,t}^2). \end{aligned}$$

Dynamic evolution: Let $\eta_{it} = \log(\sigma_{it}^2)$ and $\lambda_{it} = \log(\sigma_{it}^2)$. Then,

$$\begin{aligned} \eta_{it} &\sim N(\alpha_i + \gamma_i \eta_{i,t-1}, \xi_i^2) \\ \lambda_{jt} &\sim N(\mu_j + \phi_j \lambda_{j,t-1}, \tau_j^2). \end{aligned}$$

[5] Pitt and Shephard (1999).

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Correlated factor volatilities

Aguilar and West (2000) introduce contemporaneous covariation in the common factor log-volatilities.

Let

$$\lambda_t = (\sigma^2_{p+1,t}, \ldots, \sigma^2_{p+k,t})',$$

$$\mu = (\mu_1, \ldots, \mu_k)$$

and

$$\Phi = \text{diag}(\phi_1, \ldots, \phi_k).$$

Then

$$\lambda_t \sim N(\alpha + \phi\lambda_{t-1}, U)$$

where $U$ is a full covariance matrix.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Time-varying loadings

Lopes and Carvalho (2007) introduce time-varying loadings, $\beta_t$.

The $d = pk - k(k-1)/2$ unconstrained elements of $\beta_t$, namely

$$\beta_{21,t}, \beta_{31,t}, \ldots, \beta_{p,k,t},$$

are modeled by simple first order autoregressive models, ie.

$$\beta_{ijt} \sim N(\zeta_{ij} + \Theta_{ij}\beta_{ij,t-1}, \omega_{ij}^2)$$

for $i = 2, \ldots, p$ and $j = 1, \ldots, \min(i-1, k)$.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Daily exchange rate

Returns on weekday closing spot prices for six currencies relative to the US dollar.

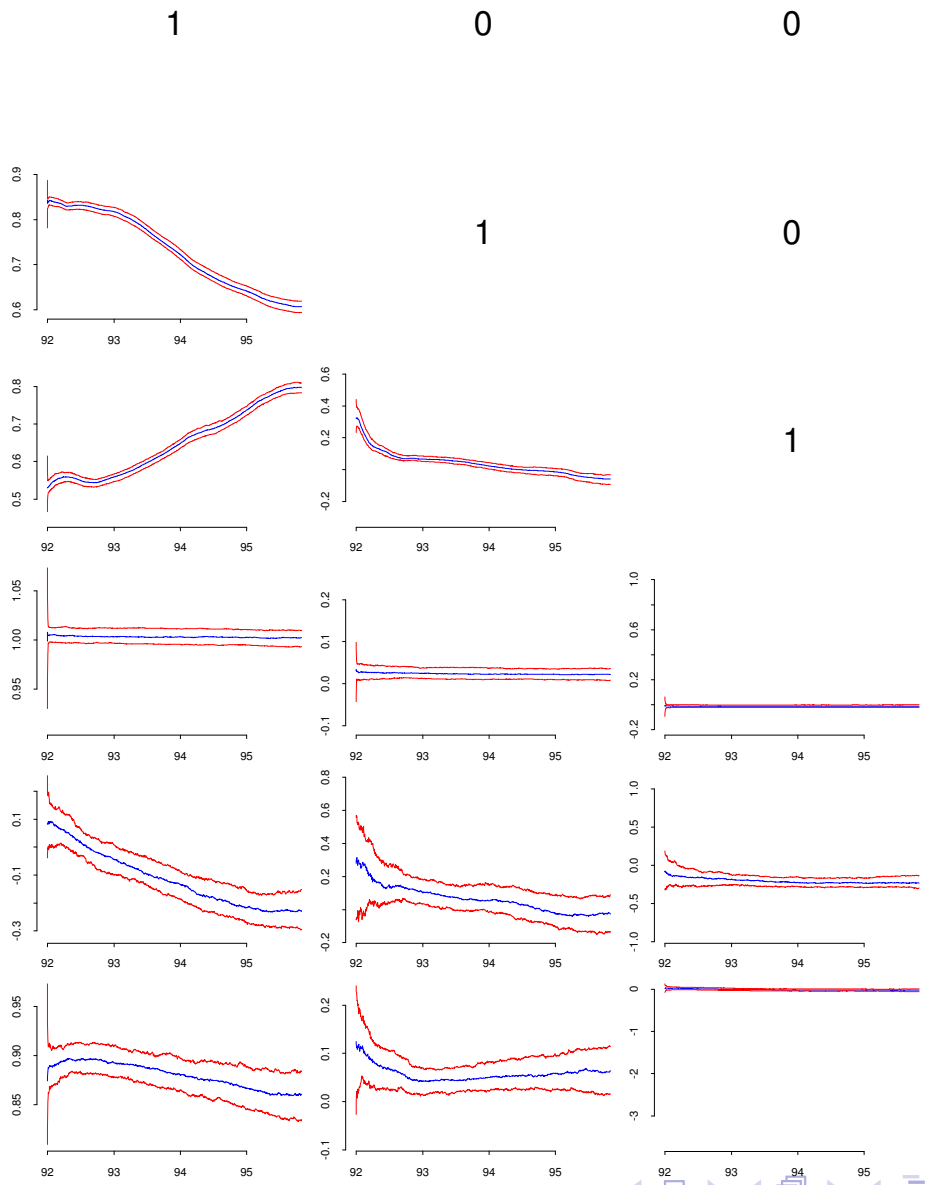
The data span the period from 1/1/1992 to 10/31/1995 inclusive.

- German Mark(DEM)
- British Pound(GBP)
- Japanese Yen(JPY)
- French Franc(FRF)
- Canadian Dollar(CAD)
- Spanish Peseta(ESP)

A $k = 3$ factor stochastic volatility model with time-varying loadings was implemented with relatively vague priors for all model parameters.

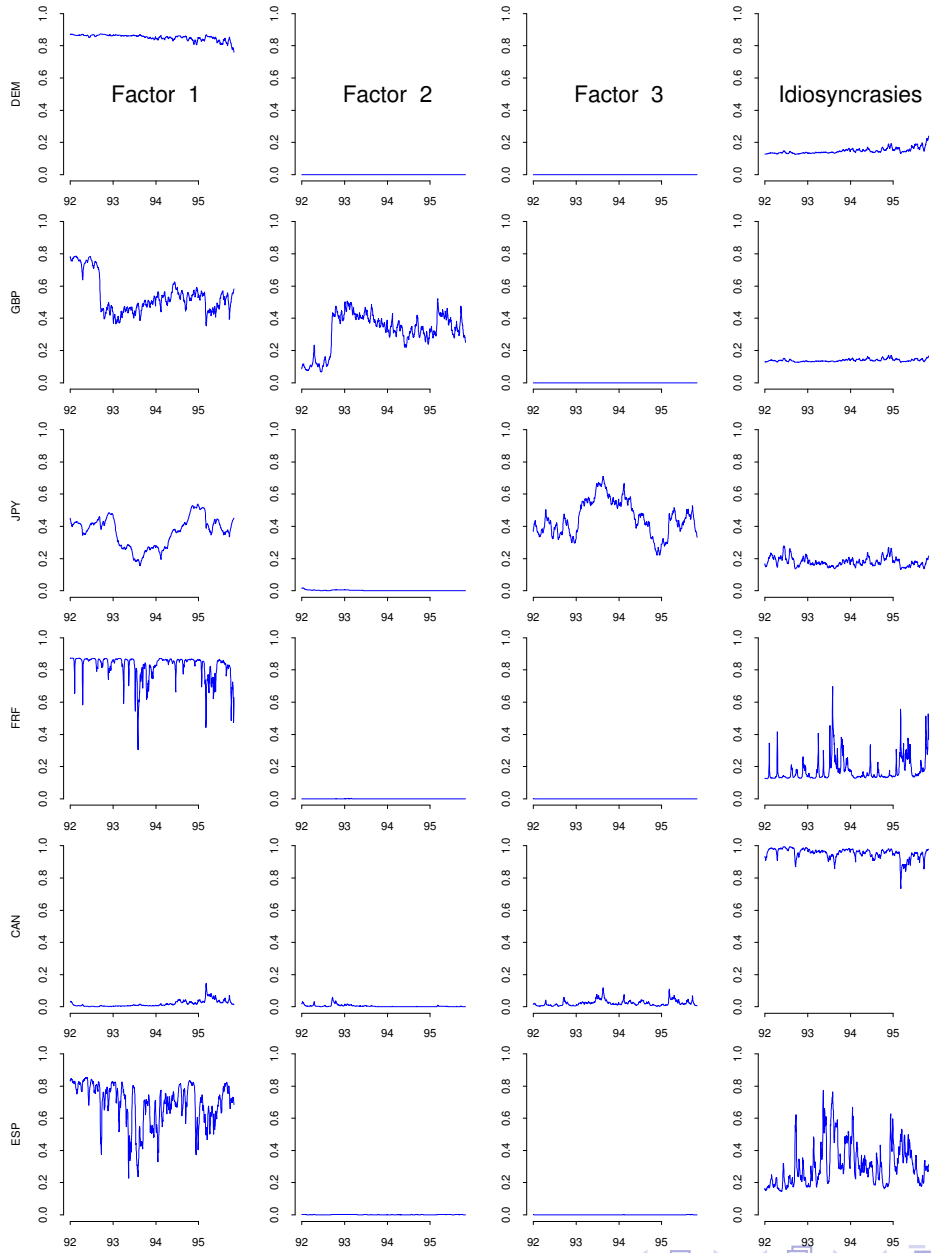
Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Factor loadings



Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Variance decomposition



Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Spatial dynamic factor models

Lopes, Salazar and Gamerman (2008) introduces the following spatio-temporal model for measurements on $m$ spatial locations and over $T$ time periods.

Factor structure:

$$y_t \sim N(\beta f_t, \Sigma)$$

Dynamic factors:

$$f_t \sim N(\Gamma f_{t-1}, \Gamma)$$

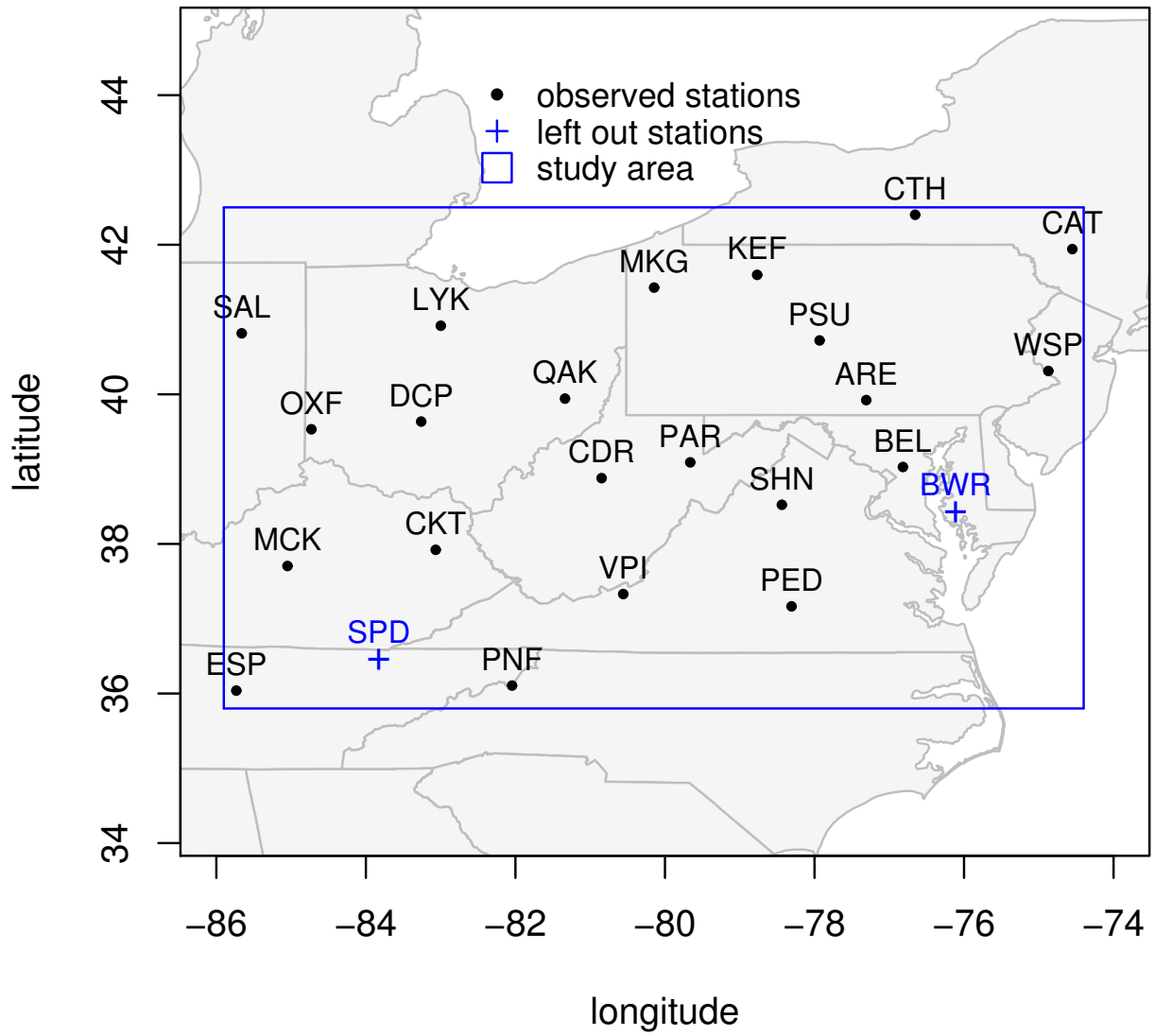
Spatial covariation:

$$\beta_j \sim GP(\mu_j, \tau_j^2 R_{\phi_j})$$

where $\beta = (\beta_1, \ldots, \beta_k)$ and $R_{\phi_j}$ spatial correlation matrix.

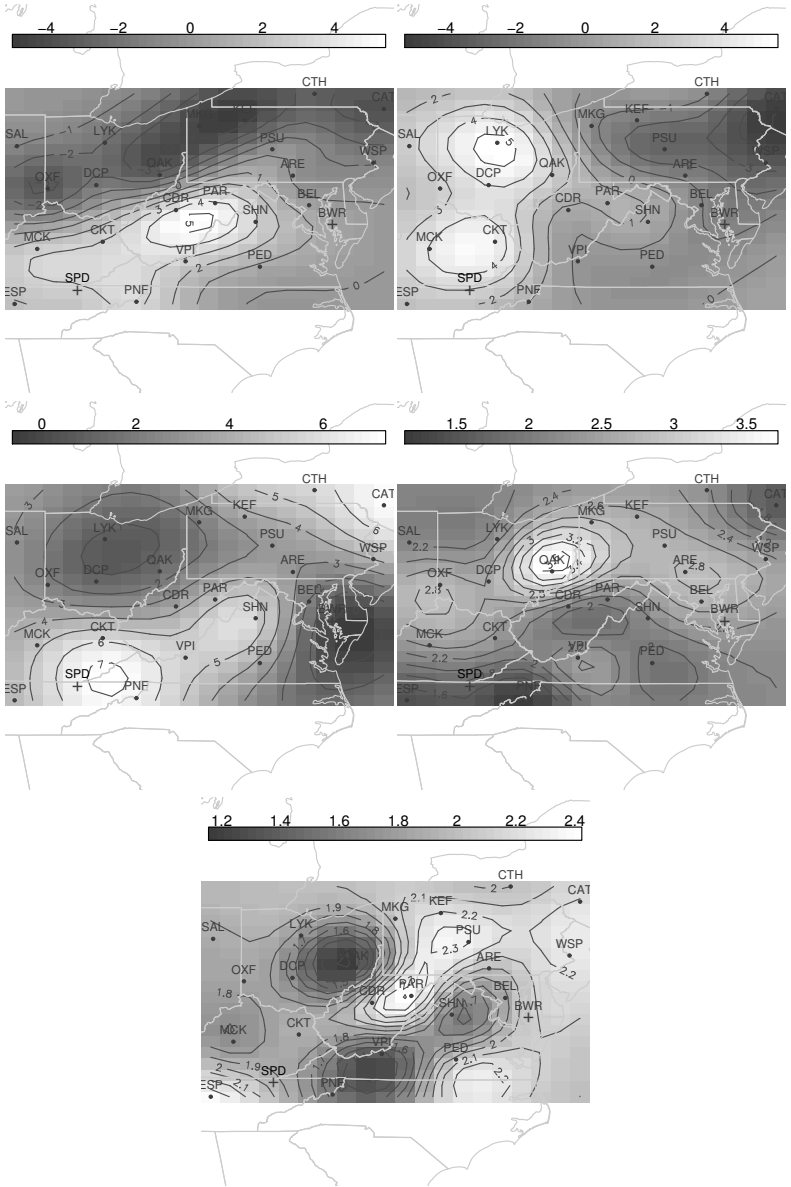
Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# $SO_2$ in the US



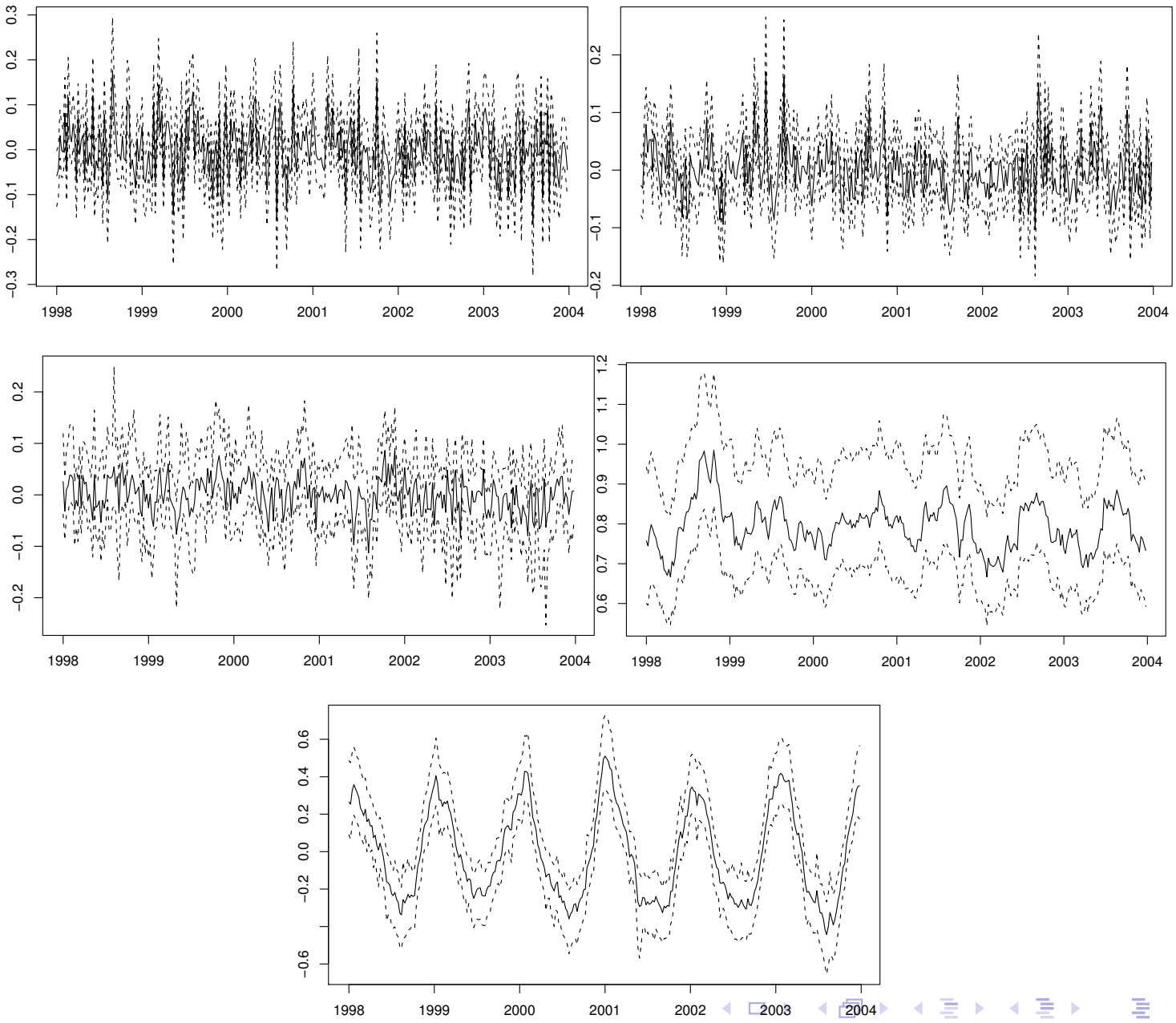
Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Spatial factor loadings



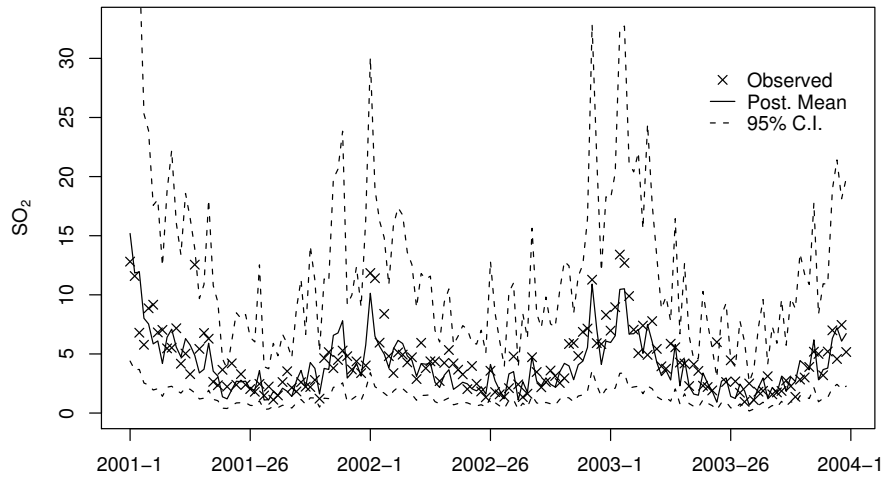
Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

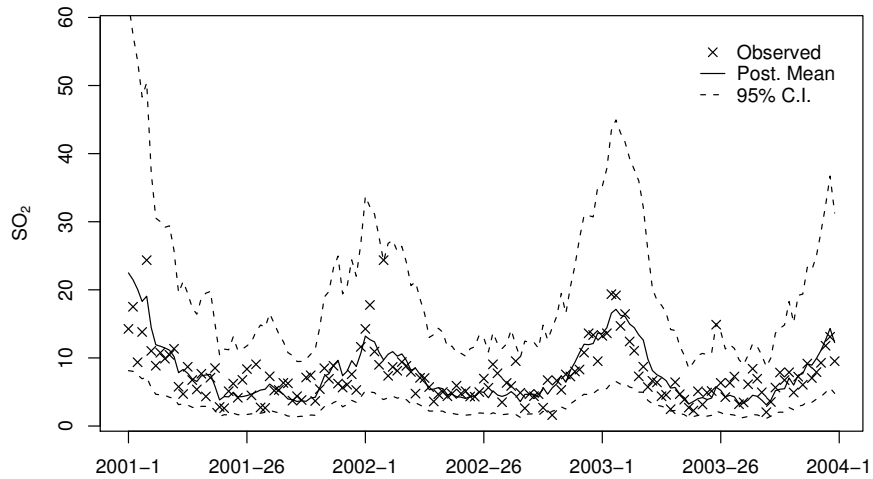
# Dynamic factors



Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Spatial interpolation

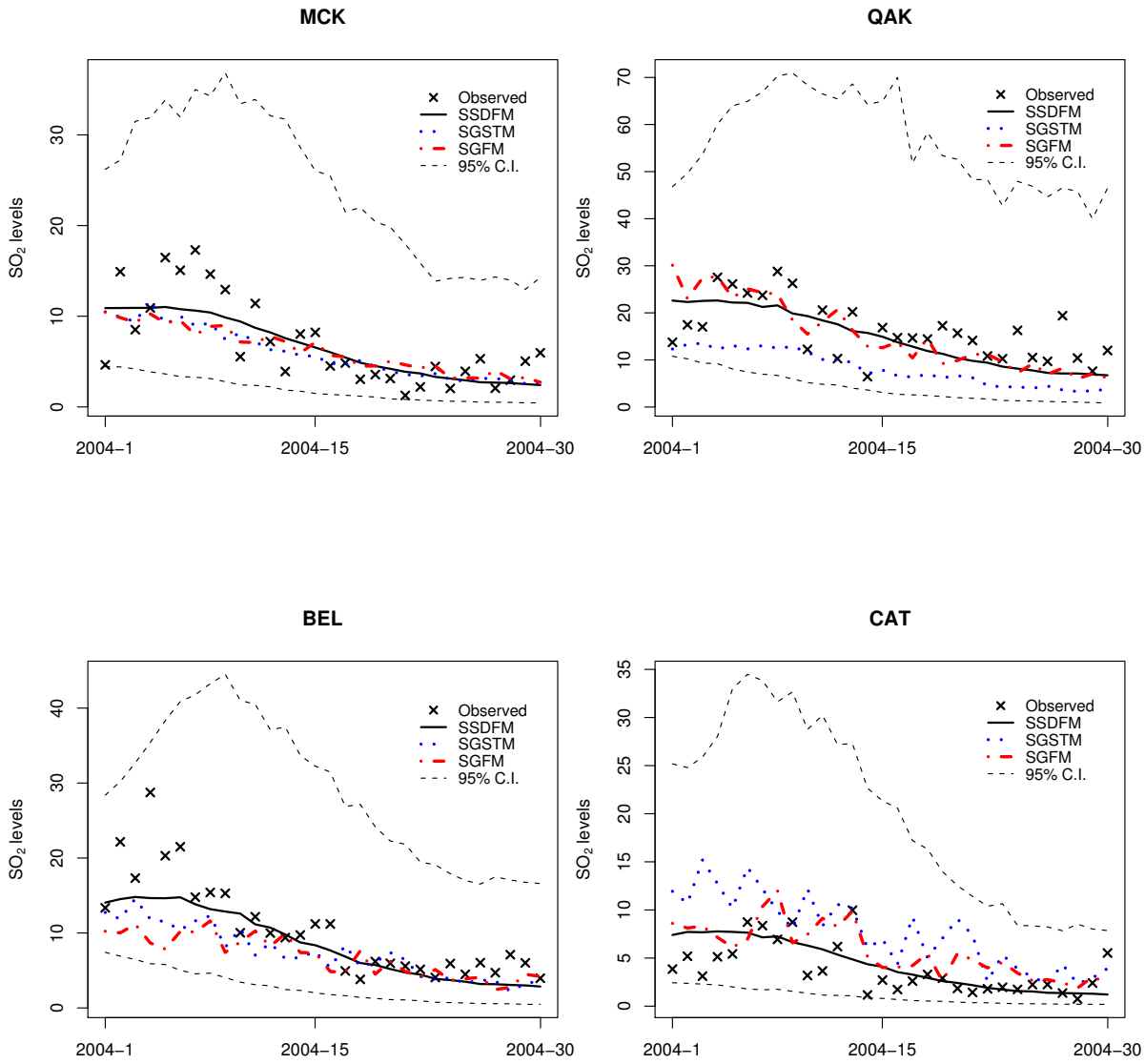



Interpolated values at stations SPD and BWR.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Forecasting



Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Forecasting with large BVAR

Carriero *et al.* (2009) forecasts $q = 33$ exchange rates, vis-a-vis the US dollar, with a Bayesian VAR model.

They find that the BVAR can lead to gains in forecast accuracy for the large majority of the exchange rates under analysis.

The forecast gains are typically in the range of 2%-3%, but in some relevant cases, such as the Euro-Dollar and the GBP-Dollar, they can reach 6%-9%.

A simple trading strategy based on the BVAR forecasts provides positive returns which are higher than those from RW forecasts.

In the post-2000 period the information in the exchange rates of emerging countries seems to matter for forecasting those of the developed countries more than the reverse.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Forecasting in VAR with many predictors

Korobilis (2008) addresses the issue of improving the forecasting performance of VARs when the set of available predictors is inconveniently large to handle with methods and diagnostics used in traditional small-scale models.

Assume we have available observations $x_t = (x_{t1}, \ldots, x_{tr})'$ on some macroeconomic quantities, where $r$ is large (in the order of hundreds).

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Factor analysis

A popular and simple method to incorporate into an econometric model all the information inherent in a large set of variables, is to reduce their dimension into a lower-dimensional vector of $k << n$ latent factors and insert these in the VAR model as explanatory variables:

$$\begin{aligned}(y_t|y_{t-1:t-p},\Theta) &\sim N\left(\sum_{i=1}^{p} B_i y_{t-i} + \sum_{i=0}^{s} A_i f_{t-i}, \Sigma\right)\\ (x_t|f_t,\Theta) &\sim N(\beta f_t, \Omega),\end{aligned}$$

where $\Theta = (B_1,\ldots,B_p,A_0,\ldots,A_s,\Sigma,\beta,\Omega)$.

This model is closely related o the factor augmented VAR (FAVAR) of Bernanke *et al.* (2005).

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Data

Korobilis (2008) uses the Stock and Watson (2005) data set, with $q = 132$ US variables (JAN60-DEC03).

real output and income
employment and hours
real retail, manufacturing, and trade sales
consumption
housing starts and sales
real inventories
orders
stock prices
exchange rates
interest rates and spreads
money and credit quantity aggregates
price indexes
average hourly earnings
miscellaneous

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# RMSFE

An appropriate common way to quantify out-of-sample forecasting performance is to compute the root mean square forecast error (RMSFE) statistic for each forecast horizon $h$:

$$\text{RMSFE}_{ij}^{h} = \sqrt{\sum_{t=1982:12}^{2003:12-h} (y_{i,t+h}^{*} - \tilde{y}_{i,t+h,j})^2}$$

where $y_{i,t+h}^{*}$ is the realized (observed) value of $y$ at time $t+h$ for the i-th series, and $\tilde{y}_{i,t+h,j}$ is the mean of the posterior predictive density at time $t+h$, for the i-th series, from the j-th forecasting model.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Relative RMFSE

Table C3. Forecast Comparison - relative RMSFE

| | PI | IP | EMP | UR | TBILL | PPI | CPI | PCED |
|---|---|---|---|---|---|---|---|---|
| BVAR with factors (Bayesian Model Averaging) | | | | | | | | |
| $h = 1$ | 0.94 | 1 | 0.9 | 0.96 | 1.08 | 0.88 | 0.95 | 1.09 |
| $h = 4$ | 1.06 | 0.96 | 0.93 | 0.94 | 0.95 | 0.92 | 1.05 | 0.94 |
| $h = 12$ | 0.97 | 0.92 | 0.99 | 1.02 | 0.98 | 0.92 | 0.95 | 0.96 |
| BVAR with factors (Model Selection) | | | | | | | | |
| $h = 1$ | 0.86 | 0.98 | 0.87 | 0.96 | 1.06 | 0.91 | 0.93 | 0.91 |
| $h = 4$ | 0.9 | 0.97 | 0.85 | 0.92 | 0.94 | 0.94 | 0.98 | 0.93 |
| $h = 12$ | 0.87 | 0.99 | 0.91 | 0.98 | 0.89 | 0.87 | 0.99 | 0.96 |
| VAR with factors (BIC Selection) | | | | | | | | |
| $h = 1$ | 0.92 | 0.99 | 0.94 | 0.99 | 1.22 | 0.99 | 1.01 | 0.97 |
| $h = 4$ | 0.93 | 0.97 | 0.94 | 0.94 | 1.12 | 0.97 | 1.06 | 0.94 |
| $h = 12$ | 0.97 | 1.04 | 0.98 | 1.05 | 0.99 | 0.9 | 1.1 | 0.95 |

Note: The variables of interest are: PI: Personal Income (A0M052), IP: Industrial Production(IPS10), EMP:Employment Rate (CES002), UR: Unemployment Rate (LHUR), TBILL: 3-month Treasury Bill Rate (FYGM3), PPI: Producer Price Index (PWFSA), CPI: Consumer Price Index (PUNEW), andPCED: PCE Deflator (GMDC)

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Large BVAR & BFAVAR

Bańbura *et al.* (2010)show that VAR with Bayesian shrinkage is an appropriate tool for large dynamic models.

When the degree of shrinkage is set in relation to the cross-sectional dimension, the forecasting performance of small monetary VARs can be improved by adding additional macroeconomic variables and sectoral information.

Large VARs with shrinkage produce credible impulse responses and are suitable for structural analysis.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Data sets

SMALL: Small monetary VAR with 3 variables.

CEE: Monetary model of Christiano et al. (1999), or SMALL + 4 variables.

MEDIUM: CEE + 13 variables.

LARGE: All 131 macroeconomic indicators from Stock and Watson (2005).

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

Table II. OLS and BVAR, relative MSFE, 1971–2003

| | | *SMALL* | | | *CEE* | | | *LARGE* |
|---|---|---|---|---|---|---|---|---|
| | | $p = 13$ | $p = \text{BIC}$ | BVAR | $p = 13$ | $p = \text{BIC}$ | BVAR | BVAR |
| $h = 1$ | EMPL | 1.14 | 0.73 | 1.14 | 7.56 | 0.76 | 0.67 | 0.46 |
| | CPI | 0.89 | 0.55 | 0.89 | 5.61 | 0.55 | 0.52 | 0.50 |
| | FFR | 1.86 | 0.99 | 1.86 | 6.39 | 1.21 | 0.89 | 0.75 |
| $h = 3$ | EMPL | 0.95 | 0.76 | 0.95 | 5.11 | 0.75 | 0.65 | 0.38 |
| | CPI | 0.66 | 0.49 | 0.66 | 4.52 | 0.45 | 0.41 | 0.40 |
| | FFR | 1.77 | 1.29 | 1.77 | 6.92 | 1.27 | 1.07 | 0.94 |
| $h = 6$ | EML | 1.11 | 0.90 | 1.11 | 7.79 | 0.78 | 0.78 | 0.50 |
| | CPI | 0.64 | 0.51 | 0.64 | 4.80 | 0.44 | 0.41 | 0.40 |
| | FFR | 2.08 | 1.51 | 2.08 | 15.9 | 1.48 | 1.30 | 1.29 |
| $h = 12$ | EMPL | 1.02 | 1.15 | 1.02 | 22.3 | 0.82 | 1.21 | 0.78 |
| | CPI | 0.83 | 0.56 | 0.83 | 21.0 | 0.53 | 0.57 | 0.44 |
| | FFR | 2.59 | 1.59 | 2.59 | 47.1 | 1.62 | 1.71 | 1.93 |

*Notes*: The table reports MSFE relative to that from the benchmark model (random walk with drift) for employment (EMPL), CPI and federal funds rate (FFR) for different forecast horizons $h$ and different models. *SMALL*, *CEE* refer to the VARs with 3 and 7 variables, respectively. Those systems are estimated by OLS with number of lags fixed to 13 or chosen by the BIC. For comparison, the results of Bayesian estimation of the two models and of the large model are also provided.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

Table III. FAVAR, relative MSFE, 1971–2003

| | | FAVAR 1 factor | | | FAVAR 3 factors | | | *LARGE* |
|---|---|---|---|---|---|---|---|---|
| | | $p = 13$ | $p =$ BIC | BVAR | $p = 13$ | $p =$ BIC | BVAR | BVAR |
| $h = 1$ | EMPL | 1.36 | 0.54 | 0.70 | 3.02 | 0.52 | 0.65 | 0.46 |
| | CPI | 1.10 | 0.57 | 0.65 | 2.39 | 0.52 | 0.58 | 0.50 |
| | FFR | 1.86 | 0.98 | 0.89 | 2.40 | 0.97 | 0.85 | 0.75 |
| $h = 3$ | EMPL | 1.13 | 0.55 | 0.68 | 2.11 | 0.50 | 0.61 | 0.38 |
| | CPI | 0.80 | 0.49 | 0.55 | 1.44 | 0.44 | 0.49 | 0.40 |
| | FFR | 1.62 | 1.12 | 1.03 | 3.08 | 1.16 | 0.99 | 0.94 |
| $h = 6$ | EMPL | 1.33 | 0.73 | 0.87 | 2.52 | 0.63 | 0.77 | 0.50 |
| | CPI | 0.74 | 0.52 | 0.55 | 1.18 | 0.46 | 0.50 | 0.40 |
| | FFR | 2.07 | 1.31 | 1.40 | 3.28 | 1.45 | 1.27 | 1.29 |
| $h = 12$ | EMPL | 1.15 | 0.98 | 0.92 | 3.16 | 0.84 | 0.83 | 0.78 |
| | CPI | 0.95 | 0.58 | 0.70 | 1.98 | 0.54 | 0.64 | 0.44 |
| | FFR | 2.69 | 1.43 | 1.93 | 7.09 | 1.46 | 1.69 | 1.93 |

*Notes*: The table reports MSFE for the FAVAR model relative to that from the benchmark model (random walk with drift) for employment (EMPL), CPI and federal funds rate (FFR) for different forecast horizons $h$. FAVAR includes 1 or 3 factors and the three variables of interest. The system is estimated by OLS with number of lags fixed to 13 or chosen by the BIC and by applying Bayesian shrinkage. For comparison the results from large Bayesian VAR are also provided.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

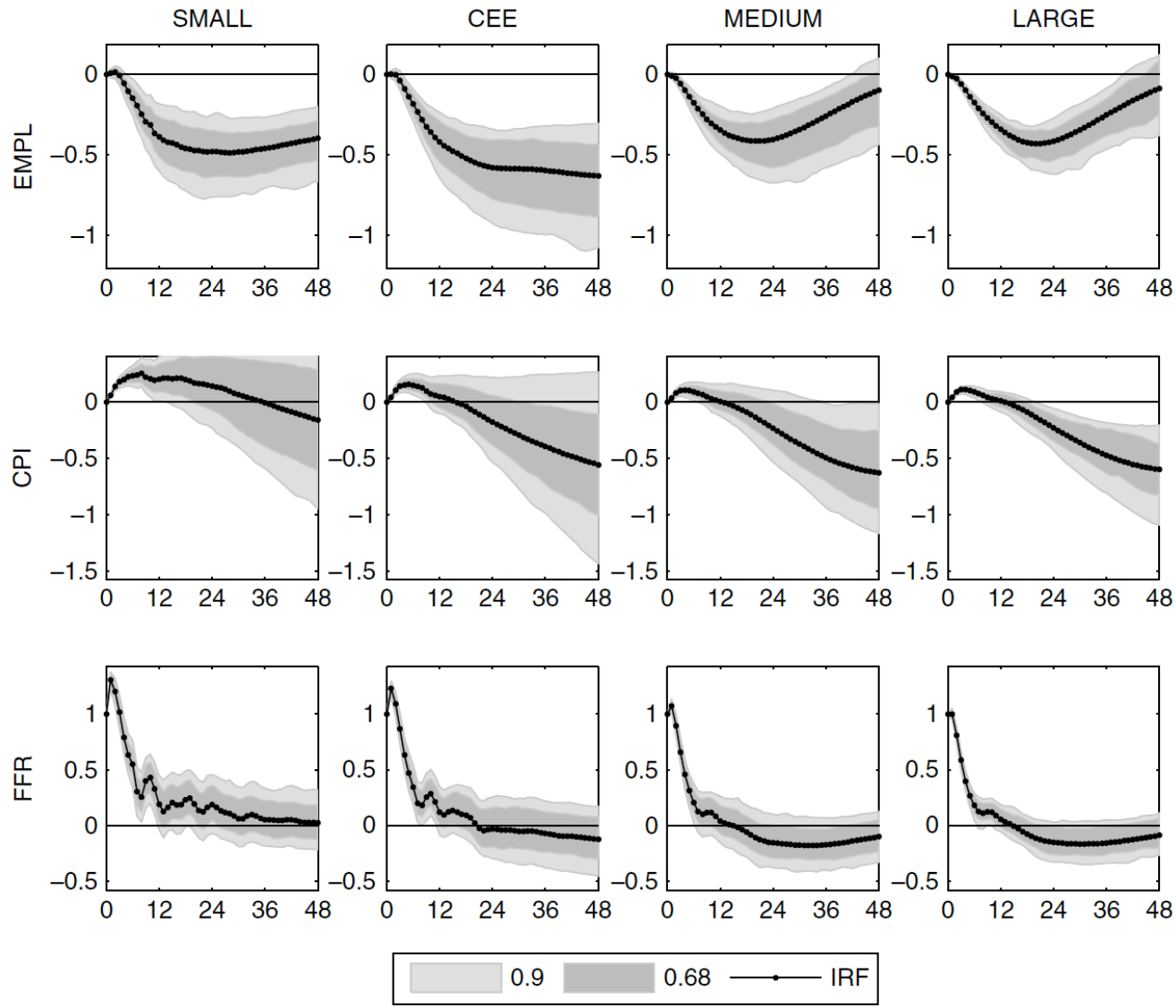



Figure 1. BVAR, impulse response functions. The figure presents the impulse response functions to a monetary policy shock and the corresponding posterior coverage intervals at 0.68 and 0.9 level for employment (EMPL), CPI and federal funds rate (FFR). *SMALL*, *CEE*, *MEDIUM* and *LARGE* refer to VARs with 3, 7, 20 and 131 variables, respectively. The prior on the sum of coefficients has been added with the hyperparameter $\tau = 10\lambda$.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Large TVP-BVAR & BFAVAR

Koop and Korobilis (2012) develop methods for estimation and forecasting in large time-varying parameter vector autoregressive models (TVP-VARs).

They use the SUR representation of the VAR(p) model with time-varying parameters

$$\begin{aligned} y_t &= z_t\beta_t + u_t \qquad & u_t \sim N(0, \Sigma) \\ \beta_t &= \beta_{t-1} + v_t \qquad & v_t \sim N(0, V_t) \end{aligned}$$

They use the old *discount* strategy of West and Harrison (1997) to avoid MCMC altogether.

See Ahmadi and Uhlig (2009) for a fully Bayesian FAVAR alternative.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# Review papers

Two recent review chapters in the same volume are

Geweke and Whiteman (2006) Bayes Forecasting.
In *Handbook of Economic Forecasting*, 3-80.

Stock and Watson (2006) Forecasting with Many Predictors.
In *Handbook of Economic Forecasting*, 515-554.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# References

Aguilar and West(2000). Bayesian dynamic factor models and variance matrix discounting for portfolio allocation. *Journal of Business and Economic Statistics*, 18, 338-357.

Ahmadi and Uhlig (2009). Measuring the Dynamic Effects of Monetary Policy Shocks: A Bayesian FAVAR Approach with Sign Restriction. Technical Report.

Bańbura, Giannone and Reichlin (2010). Large Bayesian vector auto regressions. JAE, 25, 71-92.

Bartholomew, D. J. (1981). Posterior analysis of the factor model. *The British Journal of Mathematical and Statistical Psychology*, 34, 93-99.

Bartholomew, D. J. (1987) *Latent Variable Models and Factor Analysis*. London: Charles Griffin.

Bernanke, Boivin and Eliasz (2005) Measuring monetary policy: a factor augmented autoregressive (FAVAR) approach. *Quarterly Journal of Economics*, 120, 387-422.

Brandt and Freeman (2006) Advances in Bayesian Time Series Modeling and the Study of Politics: Theory Testing, Forecasting, and Policy Analysis. *Political Analysis*, 14, 1-36.

Carriero, Kapetanios, Marcellino (2009) Forecasting exchange rates with a large Bayesian VAR. *International Journal of Forecasting*, 25, 400-417.

Del Negro and Schorfheide (2011) Bayesian Macroeconometrics. In *Handbook of Bayesian Econometrics*, Chapter 7, 293-387. Oxford University Press.

Doan, Litterman and Sims (1984) Forecasting and conditional projection using realistic prior distributions (with discussion). *Econometric Reviews*, 3, 1-144.

Garratt, Koop, Mise and Vahey (2009) Real-Time Prediction With U.K. Monetary Aggregates in the Presence of Model Uncertainty. *Journal of Business & Economic Statistics*, 27, 480-491.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# References (cont.)

George (2000) The variable selection problem. *Journal of the American Statistical Association*, 95, 1304-1308.

George, Sun and Ni (2008) Bayesian stochastic search for VAR model restrictions. *Journal of Econometrics*, 142, 553-580.

George and McCulloch (1993) Variable selection via Gibbs sampling. *Journal of the American Statistical Association*, 88, 881-889.

George and McCulloch (1997) Approaches to Bayesian variable selection. *Statistica Sinica*, 7, 339-379.

Geweke and Singleton (1980). Interpreting the likelihood ratio statistic in factor models when sample size is small. *Journal of the American Statistical Association*, 75, 133-137.

Geweke and Zhou (1996). Measuring the pricing error of the arbitrage pricing theory. *The Review of Financial Studies*, 9, 557-587.

Ihara and Kano (1995) Identifiability of full, marginal, and conditional factor analysis model. *Statistics and Probability Letters*, 23, 343-50.

Jochmann, Koop, Leon-Gonzalez and Strachan (2011) Stochastic search variable selection in vector error correction models with application to a model of the UK macroeconomy. *Journal of Applied Econometrics*.

Jochmann, Koop and Strachan (2010) Bayesian forecasting using stochastic search variable selection in a VAR subject to breaks. *International Journal of Forecasting*, 26, 326-347.

Kadiyala and Karlsson (1993) Forecasting with generalized Bayesian vector autoregressions. *Journal of Forecasting*, 12, 365-78.

Kadiyala and Karlsson (1997). Numerical methods for estimation and inference in Bayesian VAR-models. *Journal of Applied Econometrics*, 12, 99-132.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# References (cont.)

Koop and Korobilis (2009). Bayesian multivariate time series methods for empirical macroeconomics. *Foundations and Trends in Econometrics*, 3, 267-358.

Koop and Korobilis (2012) Large Time-Varying Parameter VARs. Technical Report.

Korobilis (2008) Forecasting in vector autoregressions with many predictors. In *Bayesian Econometrics (Advances in Econometrics)*, 23, 403-431.

Korobilis (2012) VAR forecasting using Bayesian variable selection. *Journal of Applied Econometrics*, forthcoming.

Litterman (1980) A Bayesian procedure for forecasting with vector autoregressions. Technical Report, MIT.

Litterman (1986) Forecasting with Bayesian vector autoregressions – five years of experience. *Journal of Business & Economic Statistics*, 4, 25-38.

Lopes (2000) Bayesian Analysis in Latent Factor and Longitudinal Models. Unpublished PhD. Thesis, Institute of Statistics and Decision Sciences, Duke University.

Lopes (2003) Factor models. *ISBA Bulletin*, 10(2), 7-10.
`http://faculty.chicagobooth.edu/hedibert.lopes/research/pdf/factormodels-lopes.pdf`

Lopes and Carvalho (2007) Factor stochastic volatility with time varying loadings and Markov switching regimes. *Journal of Statistical Planning and Inference*, 137, 3082-3091.

Lopes, McCulloch and Tsay (2012) Cholesky stochastic volatility models for high-dimensional time series. Technical Report, Chicago Booth.

Lopes and Polson (2012) Bayesian instrumental variables: likelihoods and priors. *Econometric Reviews*.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# References (cont.)

Lopes and West (2004) Bayesian model assessment in factor analysis. *Statistica Sinica*, 14, 41-67.

Lopes, Salazar and Gamerman (2008) Spatial dynamic factor models. *Bayesian Analysis*,

Lütkepohl (2007) *New Introduction to Multiple Time Series Analysis*. Springer.

Martin and McDonald (1975) Bayesian estimation in unrestricted factor analysis: A treatment for Heywood cases. *Psychometrika*, 40, 505-517.

Nakajima, Kasuya and Watanabe (2011) Bayesian analysis of time-varying parameter vector autoregressive model for the Japanese economy and monetary policy. *Journal of The Japanese and International Economies*, 25, 225-245.

Pelloni and Polasek (2003) Macroeconomic Effects of Sectoral Shocks in Germany, the U.K. and, the U.S.: A VAR-GARCH-M Approach. *Computational Economics*, 21, 65-85.
Pfaff (2008) VAR, SVAR and SVEC Models: Implementation within `R` Package `vars`. *Journal of Statistical Software*, 27, 1-32.

Pitt and Shephard (1999) Time varying covariances: A factor stochastic volatility approach. In *Bayesian statistics 6* (ed. J.M. Bernardo et al). London: Oxford University Press.

Pourahmadi (1999) Joint mean-covariance models with applications to longitudinal data: Unconstrained parameterisation. *Biometrika* 86, 677-690.

Press (1982) *Applied Multivariate Analysis: Using Bayesian and Frequentist Methods of Inference* (2nd edition). New York: Krieger.

Press and Shigemasu (1989) Bayesian inference in factor analysis. In *ASA Proceedings of Social Statistics Section*, 292-294.

Variable selection
ON/OFF switch
SSVS
Factor analysis
Basic model
PBFA
FSV
SDFM
Forecasting with large $q$
Forecasting with many predictors
Large BVAR & BFAVAR
Large TVP-BVAR & BFAVAR
References

# References (cont.)

Primiceri (2005). Time Varying Structural Vector Autoregressions and Monetary Policy, *Review of Economic Studies*, 72, 821-852.

Rubio-Ramírez, Waggoner and Zha (2010) Structural Vector Autoregressions: Theory of Identification and Algorithms for Inference. *Review of Economic Studies*, 77, 665-696.

Sims (1980) Macroeconomics and Reality. *Econometrica*, 48, 1-48.

Sims (2007) Bayesian Methods in Applied Econometrics, or, Why Econometrics Should Always and Everywhere Be Bayesian. Technical Report, Princeton University.

Sims and Zha (1998). Bayesian Methods for Dynamic Multivariate Models. *International Economic Review*, 39, 949-968.

Smith and Kohn (2002) Parsimonious covariance matrix estimation for longitudinal data. *Journal of the American Statistical Association*, 97, 1141-1153.

Stock and Watson (2005) Implications of dynamic factor models for VAR analysis. Technical Report.

Uhlig (1997) Bayesian Vector Autoregressions with Stochastic Volatility. *Econometrica*, 65, 59-73.

Villani (2009) Steady-state priors for vector autoregressions. *Journal of Applied Econometrics*, 24, 630-650.

Waggoner and Zha (2003) A Gibbs Sampler for Structural VARs. *Journal of Economic Dynamics and Control*, 28, 349-366.

West (2003) Bayesian factor regression models in the "large p, small n" paradigm". in *Bayesian Statistics 7*, 723-732.

Zha (1999) Block recursion and structural vector autoregressions. *Journal of Econometrics*, 90, 291-316.

Zivot and Wang (2003) *Modeling Financial Time Series with S-plus*. Springer.